Suraj Sah Kanu

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Professional Summary

Machine Learning Engineer and AI Researcher with a proven track record of designing, developing, and deploying enterprise-ready AI systems. Expert in architecting advanced **Retrieval-Augmented Generation (RAG)** pipelines, **Multi-Agent AI Systems** (**LangGraph, CrewAI**), and custom LLM fine-tuning workflows utilizing parameter-efficient methods (**QLoRA/LoRA**). Adept at bridging the gap between cutting-edge AI research and production environments by building scalable APIs with **FastAPI**, containerizing applications via **Docker**, and implementing robust MLOps practices on cloud platforms.

Core Competencies

Machine Learning • Deep Learning • Computer Vision • Medical Image Analysis • Transfer Learning • Model Optimization • Feature Engineering • Supervised & Unsupervised Learning • Hyperparameter Tuning • Retrieval-Augmented Generation (RAG) • Generative AI • MLOps • CI/CD • Data Augmentation • Descriptive Statistics • Inferential Statistics • A/B Testing • Hypothesis Testing

Technical Skills

Programming Languages: Python, SQL, Bash

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, YOLO, Transfer Learning, Computer Vision

LLM & Generative AI: Hugging Face Transformers, QLoRA, LoRA, PEFT, TRL, Unsloth, LangChain, LangGraph, CrewAI, Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, Model Fine-Tuning

Vector Databases Qdrant, FAISS, Pinecone, ChromaDB

Backend & Databases: FastAPI, SQLAlchemy, Redis, MySQL, PostgreSQL, SQLite

MLOps & Deployment: Docker, Kubernetes, MLflow, GitHub Actions, CI/CD, AWS SageMaker, Streamlit

Data Science & Statistics: Pandas, NumPy, Matplotlib, Plotly, Power BI, Descriptive Statistics, Inferential Statistics, A/B Testing, PCA, Regression Analysis, Probability Theory

Developer Tools: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, PyCharm, Linux, Anaconda

Experience

Machine Learning and Deep Learning Research Intern Aug 2025 – Apr 2026

NepAI Applied Mathematics and Informatics Institute for Research (NAAMII) – MAPMED Lab *Kathmandu, Nepal*

- Built end-to-end **medical image embedding extraction pipeline** processing 3D volumetric CT/MRI scans using MedicalNet ResNet models; implemented global and ROI-based embeddings across 4 feature layers (256–2048 dims).
- Engineered **3 fine-tuning strategies** (partial, full, linear probe) for automated esophagitis grading using the RTOG dataset, achieving **92%+ validation accuracy** via stratified k-fold cross-validation.
- Developed advanced **3D preprocessing pipeline** with center-crop normalization, multi-scale augmentation, and NIfTI format conversion, reducing image processing overhead by **65%** through batch automation.
- Created comprehensive **embedding analysis framework** with PCA dimensionality reduction, contrastive similarity metrics, Isolation Forest outlier detection, and 16+ automated visualizations for quality assurance.
- Designed robust data utilities including CSV label merging, DICOM/NIfTI validation, cross-platform logging, and production-ready error handling; documented **15+ technical workflows** for reproducible research.

Projects

Production-Ready Qwen3-4B LLM Fine-Tuning Pipeline | *Unsloth, Transformers, TRL, PEFT, QLoRA* Kaggle Code

- Designed and implemented an end-to-end **QLoRA fine-tuning pipeline** for **Qwen3-4B-Instruct** using **4-bit NF4 quantization**, PEFT, and Unsloth, enabling efficient supervised fine-tuning on dual NVIDIA T4 GPUs.
- Built a configurable data pipeline supporting both **human-curated** and **LLM-generated synthetic instruction datasets**, enabling reproducible experimentation across multiple training datasets.
- Optimized training with **TRL's SFTTrainer** by tuning LoRA rank, alpha, learning rate scheduling, gradient accumulation, sequence length, and mixed-precision settings for stable convergence under GPU memory constraints.
- Implemented automated experiment tracking by saving **training logs, checkpoints, evaluation metrics, generated outputs, tokenizer artifacts, and configuration files**, ensuring reproducibility and efficient model comparison.
- Developed an evaluation and inference pipeline to compare base and fine-tuned models through qualitative response analysis, validating instruction-following improvements before deployment.
- Exported trained models as **LoRA adapters, merged FP16 checkpoints, and GGUF** for deployment on Hugging Face Transformers, Ollama, and llama.cpp; published the complete notebook, model, and training artifacts on **Kaggle.**

AI-Powered Multi-Agent Hotel Booking Assistant | *Python, CrewAI, NVIDIA NIM, LiteLLM, JSONC* github

- Developed an intelligent **multi-agent hotel booking system** using CrewAI, where specialized AI agents collaboratively handled travel consultation, hotel search, and booking assistance through sequential task orchestration.
- Designed autonomous agents to analyze customer travel preferences, including destination, budget, travel dates, hotel type, amenities, and special requests, enabling personalized hotel recommendations.

- Integrated **NVIDIA NIM Large Language Models** via LiteLLM to power context-aware reasoning, natural language understanding, and agent-to-agent collaboration throughout the booking workflow.
- Implemented context sharing and sequential execution between agents, allowing seamless information flow from requirement gathering to hotel recommendation and booking summary generation.
- Built a modular and extensible architecture supporting future integration with hotel search APIs, Google Maps, live pricing, payment gateways, RAG knowledge bases, and Streamlit-based user interfaces.
- Structured the project using configurable JSONC agent definitions, environment-based configuration, and reusable workflows to facilitate rapid customization and scalable deployment.

Neural RAG API Conversational AI | *FastAPI, LangChain, NVIDIA NIM, Qdrant, Redis, SQLAlchemy, Docker* github

- Developed a production-ready **Retrieval-Augmented Generation (RAG)** backend enabling document-aware conversational AI with semantic retrieval, reducing LLM hallucinations by **up to 95%** through retrieval-grounded responses and context-aware generation.
- Built a custom RAG pipeline integrating **Qdrant Vector Database**, **NVIDIA embedding models**, and **LangChain**, delivering low-latency semantic search with citation-grounded responses.
- Implemented persistent multi-turn conversational memory using **Redis** with automatic session management and configurable TTL for enterprise-scale chat applications.
- Designed an intelligent document ingestion pipeline supporting PDF processing, semantic chunking, embedding generation, and scalable vector indexing for efficient knowledge retrieval.
- Engineered an LLM-powered interview booking extraction service that automatically extracts structured candidate information (name, email, interview date, and time) from natural language conversations into a SQL database.
- Containerized the application using **Docker Compose** and designed a modular service-oriented architecture with environment-based configuration for scalable production deployment.

Agentic AI Compliance Intelligence System | *FastAPI, LangGraph, OpenAI GPT-4o, Pydantic* github

- Architected and deployed a **multi-agent document intelligence system** using LangGraph to transform unstructured manufacturer PDFs into structured, audit-ready compliance reports for **Nepal import review workflows.**
- Designed a **5-stage agentic pipeline** (extraction → normalization → reconciliation → compliance mapping → report generation), enabling end-to-end automated reasoning over heterogeneous technical documents.
- Built a GPT-4o powered **cross-document reconciliation engine** that detects specification conflicts, aligns variant-level inconsistencies, and assigns **confidence-scored validation outputs** across multiple PDF sources.
- Implemented a strongly-typed extraction layer using **Pydantic v2 schemas**, ensuring deterministic structuring of product, manufacturer, certification, and technical specification data from noisy unstructured PDFs.
- Engineered a hybrid PDF ingestion system using **PyMuPDF + pdfplumber**, improving extraction robustness for complex layouts including tables, mixed formatting, and semi-scanned industrial documents.
- Introduced a **full source-traceability framework**, linking every extracted field back to originating documents, enabling auditability and compliance transparency for regulatory-grade workflows.
- Generated structured outputs including **JSON extraction artifacts, reconciliation reports, and compliance-grade Markdown documents**, aligned with NEPQA 2025 import-side review expectations.
- Delivered a production-ready backend with **FastAPI + CLI orchestration**, supporting reproducible execution, modular service design, structured logging, and failure-resilient agent workflows.

Education

Bachelor in Computer Application
National Infotech College, Tribhuvan University — *Nepal*

Higher Secondary Education (Science)
Birgunj Public College — *Birgunj, Nepal*

Achievements & Certifications

AWS AI/ML Scholar 2025 – Selected among **top 3%** of applicants through competitive AWS AI & ML Scholarship Program based on technical excellence.

AWS Certified: Machine Learning Foundations – Hands-on expertise in ML principles, AWS services, and cloud-based model training.

AWS Certified: Generative AI with AWS – Practical experience with Amazon Bedrock, foundation models, and generative AI application development.

Positions of Responsibility

Teaching Assistant 2025
Annual Nepal AI School — *Nepal*

- Supported instructors in delivering **AI and data science** sessions; assisted students with hands-on labs and doubt resolution, evaluated assignments, and coordinated workshops and technical events.